

Now, $z_1 = \frac{(4,2)}{(3,-1)} = \frac{4+2i}{3-i}$

Multiply the numerator and denominator by the complex conjugate of $z_2 = 3-i$.

$z_1 = \frac{4+2i}{3-i} \times \frac{3+i}{3+i} = \frac{(4+2i)(3+i)}{(3-i)(3+i)}$

$= \frac{12+4i+6i+2i^2}{9-i^2} = \frac{12+10i-2}{9-(-1)} = \frac{10+10i}{10} = 1+i$

$\therefore \bar{z} = -1$

Thus, $z_1 = 1+i$

Modulus of Complex Number: The real number $\sqrt{x^2 + y^2}$ is called the modulus of the complex number $x+iy$ and it is denoted by $|x+iy|$. In Figure, $|OA|$ represent the modulus of $x+iy$. OR In other words, the modulus of a complex number is the distance from the origin to the point representing the number.



Example 3: If $z = \frac{(1+2i)^2}{2-i}$ then evaluate $|\bar{z}|$

Sol. $z = \frac{(1+2i)^2}{2-i} = \frac{1+4i+4i^2}{2-i} = \frac{-3+4i}{2-i}$

$= \frac{-3+4i}{2-i} \times \frac{2+i}{2+i} = \frac{-6-3i+8i+4i^2}{2^2-i^2} = \frac{-6+5i-4}{4-(-1)} = \frac{-10+5i}{5} = -2+i$

$\Rightarrow z = -2+i$

Taking conjugate $\bar{z} = -2-i$

and $|\bar{z}| = |-2-i| = \sqrt{(-2)^2 + (-1)^2} = \sqrt{4+1} = \sqrt{5}$

Note:

- Every real number is a complex number with 0 as its imaginary part.
- A real number is self-conjugate.
- The set C of complex numbers does not satisfy the order axioms. In fact, there is no sense in saying that one complex number is greater or less than the other.

Exercise 1.1

Q1. Find the multiplicative inverse of each of the following numbers:

(i) $(-4, 7)$

Sol. Given that $z = (-4, 7)$

$\Rightarrow z = -4+7i$

So, $\frac{1}{z} = \frac{1}{-4+7i} \Rightarrow \frac{1}{z} \times \frac{-4-7i}{-4-7i} = \frac{-4-7i}{16-49i^2}$

$= \frac{-4-7i}{16+49} = \frac{-4-7i}{65} = -\frac{4}{65} - \frac{7}{65}i$

$\Rightarrow \frac{1}{z} = -\frac{4}{65} - \frac{7}{65}i$

(ii) $(\sqrt{2}, -\sqrt{5})$

Sol. Given that $z = (\sqrt{2}, -\sqrt{5})$

$\Rightarrow z = \sqrt{2}-\sqrt{5}i$

So, $\frac{1}{z} = \frac{1}{\sqrt{2}-\sqrt{5}i} \Rightarrow \frac{1}{z} \times \frac{\sqrt{2}+\sqrt{5}i}{\sqrt{2}+\sqrt{5}i} = \frac{\sqrt{2}+\sqrt{5}i}{2-5i^2}$

$= \frac{\sqrt{2}+\sqrt{5}i}{2+5} = \frac{\sqrt{2}+\sqrt{5}i}{7}$

$\Rightarrow \frac{1}{z} = \frac{\sqrt{2}}{7} + \frac{\sqrt{5}}{7}i$

(iii) $(1, 0)$

Sol. Given that $z = (1, 0)$

$\Rightarrow z = 1+0i$

$\Rightarrow z = 1$

So, $\frac{1}{z} = 1 \Rightarrow \frac{1}{z} = (1, 0)$

Q2. Separate into real and imaginary parts (write as a simple complex number):

(i) $\frac{2-7i}{4+5i}$

Sol. Let $z = \frac{2-7i}{4+5i}$

$= \frac{2-7i}{4+5i} \times \frac{4-5i}{4-5i} = \frac{2(4-5i)-7i(4-5i)}{(4)^2-(5i)^2}$

$= \frac{8-10i-28i+35i^2}{16-25i^2} = \frac{8-10i-28i+35(-1)}{16-25(-1)} = \frac{8-10i-28i-35}{16+25}$

$= \frac{-27-38i}{41} = -\frac{27}{41} - \frac{38}{41}i$

So, $\text{Re}(z) = -\frac{27}{41}$ and $\text{Im}(z) = -\frac{38}{41}$

(ii) $\frac{(-2+3i)^2}{1+i}$

Sol. let $z = \frac{(-2+3i)^2}{1+i}$

$= \frac{(-2)^2 + (3i)^2 + 2(-2)(3i)}{1+i} = \frac{4-9i^2-12i}{1+i} = \frac{4+9-12i}{1+i}$

$= \frac{13-12i}{1+i} \times \frac{1-i}{1-i} = \frac{(13-12i)(1-i)}{1-i^2} = \frac{13-12i-12i+12i^2}{1-(-1)}$

$= \frac{13-24i-12}{2} = \frac{1-24i}{2} = \frac{1}{2} - 12i$

So, $\text{Re}(z) = \frac{1}{2}$ and $\text{Im}(z) = -12$

Q3. Prove that $\bar{\bar{z}} = z$ iff z is real.

Sol. Assume that $z = a+bi$

then $\bar{z} = a-bi$

Given $\bar{\bar{z}} = z$

$a-bi = a+bi$

$a-bi-a-bi=0$

$-2bi=0 \Rightarrow b=0$

So, $z = a+0i = a$

Hence z is a real number if $\bar{\bar{z}} = z$.

Conversely,

Assume that z is a real number.

$z = a \Rightarrow \bar{z} = a = a+0i$

Hence $\bar{\bar{z}} = z$ if z is a real number.

Q4. For $z \in \mathbb{C}$, show that

(i) $\frac{z+\bar{z}}{2} = \text{Re}(z)$

Sol. Take, L.H.S. $= \frac{z+\bar{z}}{2}$

Assume that $z = a+bi$

So, L.H.S. $= \frac{(a+bi) + (\overline{a+bi})}{2}$

$= \frac{(a+bi) + (a-bi)}{2} = \frac{a+bi+a-bi}{2} = \frac{2a}{2} = a$

L.H.S. $= \text{Re}(z)$

L.H.S. $= \text{R.H.S}$

$$x + iy + 2 - 3i = i[15 + 20i - 3i - 4(-1)]$$

$$x + iy + 2 - 3i = i[15 + 20i - 3i + 4]$$

$$x + iy + 2 - 3i = i(19 + 17i)$$

$$x + iy + 2 - 3i = 19i + 17i^2$$

$$x + iy + 2 - 3i = 19i + 17(-1)$$

$$x + iy + 2 - 3i = 19i - 17$$

$$x + iy = 19i - 17 - 2 + 3i$$

$$x + iy = -19 + 22i$$

By comparing the real and imaginary parts

$$x = -19 \text{ and } y = 22$$

$$(ii) (x + iy)(1 - i) = (2 - 3i)(-5 + 5i)(-i\frac{3}{5})$$

$$\text{Sol. } (x + iy)(1 - i) = [2(-5 + 5i) - 3(-5 + 5i)](-i\frac{3}{5})$$

$$(x + iy)(1 - i) = [-10 + 10i + 15i - 15i^2](-i\frac{3}{5})$$

$$(x + iy)(1 - i) = [-10 + 10i + 15i - 15(-1)](-i\frac{3}{5})$$

$$(x + iy)(1 - i) = [-10 + 10i + 15i - 15](-i\frac{3}{5})$$

$$(x + iy)(1 - i) = (5 + 25i)(-i\frac{3}{5})$$

$$(x + iy)(1 - i) = -3i - 15i^2$$

$$(x + iy)(1 - i) = -3i - 15(-1)$$

$$(x + iy)(1 - i) = -3i + 15$$

$$x + iy = \frac{15 - 3i}{1 - i}$$

$$x + iy = \frac{15 - 3i}{1 - i} \times \frac{1 + i}{1 + i}$$

$$x + iy = \frac{15 + 15i - 3i - 3i^2}{1^2 - i^2}$$

$$x + iy = \frac{15 + 15i - 3i - 3(-1)}{1 - (-1)}$$

$$x + iy = \frac{15 + 15i - 3i + 3}{2}$$

$$x + iy = \frac{18 + 12i}{2}$$

$$x + iy = 9 + 6i$$

$$x = 9 \text{ and } y = 6$$

By comparing the real and imaginary parts

$$x = 9 \text{ and } y = 6$$

$$(iii) \frac{x}{2+i} + \frac{y}{3-i} = 4 + 5i$$

$$\text{Sol. } \frac{(2+i)(3-i)}{(2+i)(3-i)} = 4 + 5i$$

$$\frac{3x - xi + 2y + yi}{6 - 2i + 3i - i^2} = 4 + 5i$$

$$\frac{3x + 2y - xi + yi}{6 - 2i + 3i - (-1)} = 4 + 5i$$

$$\frac{3x + 2y - xi + yi}{6 - 2i + 3i + 1} = 4 + 5i$$

$$\frac{3x + 2y - xi + yi}{7 + i} = 4 + 5i$$

$$3x + 2y - (x - y)i = (4 + 5i)(7 + i)$$

$$3x + 2y - (x - y)i = 28 + 4i + 35i + 5i^2$$

$$3x + 2y - (x - y)i = 28 + 4i + 35i + 5(-1)$$

$$3x + 2y - (x - y)i = 28 + 4i + 35i - 5$$

$$3x + 2y - (x - y)i = 23 + 39i$$

$$\text{By comparing the real and imaginary parts}$$

$$3x + 2y = 23 \quad (i)$$

$$-(x - y) = 39 \quad (ii)$$

$$\text{From (ii)}$$

$$-(x - y) = 39$$

$$-x + y = 39$$

$$y = 39 + x \quad (iii)$$

$$\text{Put in (i)}$$

$$3x + 2y = 23$$

$$3x + 2(39 + x) = 23$$

$$3x + 78 + 2x = 23$$

$$5x = 23 - 78$$

$$5x = -55$$

$$x = -\frac{55}{5}$$

$$x = -11$$

$$\text{Put in (iii)}$$

$$y = 39 + x$$

$$y = 39 - 11$$

$$y = 28$$

Q2. If $z_1 = -13 + 24i$ and $z_2 = x + yi$, find the values of x and y such that $z_1 - z_2 = -27 + 15i$

$$\text{Sol. } (-13 + 24i) - (x + yi) = -27 + 15i$$

$$-13 + 24i - x - yi = -27 + 15i$$

$$-13 + 24i + 27 - 15i = x + yi$$

$$14 + 9i = x + yi$$

By comparing the real and imaginary parts

$$x = 14 \text{ and } y = 9$$

Q3. Find the real value of x and y if:

$$(i) (x + iy)^2 = 25 + 60i$$

$$\text{Sol. } \sqrt{(x + iy)^2} = \sqrt{25 + 60i}$$

$$x + iy = \sqrt{25 + 60i}$$

By applying the square root formula for complex numbers

$$x + iy = \pm \left(\sqrt{\frac{25^2 + 60^2 + 25}{2}} + i \frac{60}{\sqrt{2}} \right)$$

$$\therefore \sqrt{x + yi} = \pm \left(\sqrt{\frac{x^2 + y^2 + x}{2}} + i \frac{y}{\sqrt{2}} \right)$$

$$x + iy = \pm \left(\sqrt{\frac{625 + 3600 + 25}{2}} + i \frac{60}{\sqrt{2}} \right)$$

$$x + iy = \pm \left(\sqrt{\frac{4225 + 25}{2}} + i \frac{\sqrt{4225 - 25}}{2} \right)$$

$$x + iy = \pm \left(\sqrt{\frac{65 + 25}{2}} + i \frac{65 - 25}{2} \right)$$

$$x + iy = \pm \left(\sqrt{\frac{90}{2}} + i \sqrt{\frac{40}{2}} \right)$$

$$x + iy = \pm \left(\sqrt{45} + i\sqrt{20} \right)$$

$$x + iy = \pm \left(3\sqrt{5} + i2\sqrt{5} \right)$$

$$\text{Either}$$

$$x + iy = + (3\sqrt{5} + i2\sqrt{5}) \text{ or } x + iy = - (3\sqrt{5} + i2\sqrt{5})$$

$$x + iy = 3\sqrt{5} + i2\sqrt{5} \text{ or } x + iy = -3\sqrt{5} - i2\sqrt{5}$$

$$\text{By comparing the real and imaginary parts}$$

$$x = 3\sqrt{5} \text{ and } y = 2\sqrt{5} \text{ or } x = -3\sqrt{5} \text{ and } y = -2\sqrt{5}$$

$$(ii) (x + iy)^2 = 64 + 48i$$

$$\text{Sol. } \sqrt{(x + iy)^2} = \sqrt{64 + 48i}$$

$$x + iy = \sqrt{64 + 48i}$$

$$\text{By applying the square root formula for complex numbers}$$

$$\therefore \sqrt{x + yi} = \pm \left(\sqrt{\frac{x^2 + y^2 + x}{2}} + i \frac{y}{\sqrt{2}} \right)$$

$$x + iy = \pm \left(\sqrt{\frac{64^2 + 48^2 + 64}{2}} + i \frac{48}{\sqrt{2}} \right)$$

$$\therefore \sqrt{x + yi} = \pm \left(\sqrt{\frac{4096 + 2304 + 64}{2}} + i \frac{48}{\sqrt{2}} \right)$$

$$x + iy = \pm \left(\sqrt{\frac{6400 + 64}{2}} + i \frac{\sqrt{6400 - 64}}{2} \right)$$

$$x + iy = \pm \left(\sqrt{\frac{80 + 64}{2}} + i \frac{\sqrt{80 - 64}}{2} \right)$$

$$x + iy = \pm \left(\sqrt{\frac{144}{2}} + i \frac{\sqrt{16}}{2} \right)$$

$$x + iy = \pm \left(\sqrt{72} + i\sqrt{8} \right)$$

$$x + iy = \pm \left(6\sqrt{2} + i2\sqrt{2} \right)$$

$$\text{Either}$$

$$x + iy = + (6\sqrt{2} + i2\sqrt{2}) \text{ or } x + iy = - (6\sqrt{2} + i2\sqrt{2})$$

$$x + iy = 6\sqrt{2} + i2\sqrt{2} \text{ or } x + iy = -6\sqrt{2} - i2\sqrt{2}$$

$$\text{By comparing the real and imaginary parts}$$

$$x = 6\sqrt{2} \text{ and } y = 2\sqrt{2} \text{ or } x = -6\sqrt{2} \text{ and } y = -2\sqrt{2}$$

$$(iii) (x + iy)^2 = \frac{2i - 3}{3 + i}$$

$$\text{Sol. } \sqrt{(x + iy)^2} = \sqrt{\frac{2i - 3}{3 + i}}$$

$$x + iy = \sqrt{\frac{2i - 3}{3 + i}} = \sqrt{\frac{(2i - 3)(3 - i)}{(3 + i)(3 - i)}}$$

$$= \sqrt{\frac{(2i - 3)(3 - i)}{9 + 1}} = \sqrt{\frac{2i(3 - i) - 3(3 - i)}{10}}$$

$$= \sqrt{\frac{6i - 2i^2 - 9 + 3i}{10}} = \sqrt{\frac{6i + 2 - 9 + 3i}{10}}$$

$$= \sqrt{\frac{-7 + 9i}{10}} = \sqrt{\frac{-7}{10} + i \frac{9}{10}}$$

$$\text{By applying the square root formula for complex numbers}$$

$$x + iy = \pm \left(\sqrt{\frac{(-7)^2 + (9)^2}{10^2}} + i \frac{9}{\sqrt{10}} \right)$$

$$= \pm \left(\sqrt{\frac{49 + 81}{100}} + i \frac{9}{\sqrt{10}} \right)$$

$$= \pm \left(\sqrt{\frac{130}{100}} + i \frac{9}{\sqrt{10}} \right)$$

$$= \pm \sqrt{\frac{\sqrt{130-7}}{2} + i \sqrt{\frac{\sqrt{130-7}}{2} + \frac{\sqrt{130-7}}{20}}}$$

$$= \pm \sqrt{\frac{\sqrt{130-7}}{2} + i \sqrt{\frac{\sqrt{130-7}}{2} + \frac{\sqrt{130-7}}{20}}}$$

$$= \pm \sqrt{\frac{\sqrt{130-7}}{10 \times 2} + i \sqrt{\frac{\sqrt{130-7}}{10 \times 2}}}$$

$$= \pm \sqrt{\frac{\sqrt{130-7}}{20} + i \sqrt{\frac{\sqrt{130-7}}{20}}}$$

Either

$$x+iy = + \left(\sqrt{\frac{\sqrt{130-7}}{20}} + i \sqrt{\frac{\sqrt{130-7}}{20}} \right)$$

or

$$x+iy = - \left(\sqrt{\frac{\sqrt{130-7}}{20}} + i \sqrt{\frac{\sqrt{130-7}}{20}} \right)$$

By comparing the real and imaginary parts

$$x = \sqrt{\frac{\sqrt{130-7}}{20}} \quad \text{and} \quad y = \sqrt{\frac{\sqrt{130-7}}{20}}$$

$$\text{or} \quad x = -\sqrt{\frac{\sqrt{130-7}}{20}} \quad \text{and} \quad y = -\sqrt{\frac{\sqrt{130-7}}{20}}$$

14. If $z_1 = 2 + 3i$ and $z_2 = 1 - \alpha$, find the value of α such that $\text{Im}(z_1 z_2) = 7$.

Sol. For $z_1 z_2$

$$z_1 z_2 = (2 + 3i)(1 - \alpha)$$

$$z_1 z_2 = 2 - 2\alpha + 3i - 3\alpha i$$

$$z_1 z_2 = (2 - 2\alpha) + (3 - 3\alpha)i$$

$$\text{Re}(z_1 z_2) = 2 - 2\alpha \quad \text{and} \quad \text{Im}(z_1 z_2) = 3 - 3\alpha$$

$$\text{Given } \text{Im}(z_1 z_2) = 7$$

$$\text{Put the value of } \text{Im}(z_1 z_2)$$

$$3 - 3\alpha = 7$$

$$-3\alpha = 7 - 3$$

$$-3\alpha = 4$$

$$\alpha = -\frac{4}{3}$$

$$\alpha = -\frac{4}{3}$$

05. If $z_1 = x + yi$ and $z_2 = a + bi$, find xy/a and b such that $z_1 + z_2 = 10 + 4i$ and $z_1 - z_2 = 6 + 2i$.

Sol. Given that $z_1 + z_2 = 10 + 4i$

$$x + yi + a + bi = 10 + 4i$$

$$x + a + yi + bi = 10 + 4i$$

$$(x + a) + (y + b)i = 10 + 4i$$

By comparing the real and imaginary parts

$$x + a = 10 \quad \text{(i)}$$

$$y + b = 4 \quad \text{(ii)}$$

Also

$$z_1 - z_2 = 6 + 2i$$

$$x + yi - (a + bi) = 6 + 2i$$

$$x + yi - a - bi = 6 + 2i$$

$$x - a + yi - bi = 6 + 2i$$

$$(x - a) + (y - b)i = 6 + 2i$$

By comparing the real and imaginary parts

$$x - a = 6 \quad \text{(i)}$$

$$y - b = 2 \quad \text{(ii)}$$

Add (i) and (ii)

$$x + a + x - a = 10 + 6$$

$$2x = 16 \Rightarrow x = \frac{16}{2} \Rightarrow x = 8$$

Put in (i)

$$x + a = 10$$

$$8 + a = 10$$

$$a = 10 - 8$$

$$a = 2$$

Add (ii) and (iv)

$$y + b + y - b = 4 + 2$$

$$2y = 6 \Rightarrow y = \frac{6}{2} \Rightarrow y = 3$$

Put in (ii)

$$y + b = 4$$

$$3 + b = 4$$

$$b = 4 - 3$$

$$b = 1$$

06. Show that $\forall z_1, z_2 \in \mathbb{C}$, $\overline{z_1 z_2} = \overline{z_1} \overline{z_2}$

Sol. Assume that $z_1 = a_1 + b_1 i$ and $z_2 = a_2 + b_2 i$

$$\text{L.H.S} = \overline{z_1 z_2}$$

$$= \overline{(a_1 + b_1 i)(a_2 + b_2 i)}$$

$$= \overline{a_1 a_2 + a_1 b_2 i + a_2 b_1 i + b_1 b_2 i^2}$$

$$= \overline{a_1 a_2 + a_1 b_2 i + a_2 b_1 i + b_1 b_2 (-1)}$$

$$= \overline{a_1 a_2 + a_1 b_2 i + a_2 b_1 i - b_1 b_2}$$

$$= \overline{a_1 a_2 - b_1 b_2 + a_1 b_2 i + a_2 b_1 i}$$

$$= (a_1 a_2 - b_1 b_2) + (a_1 b_2 + a_2 b_1)i$$

$$= (a_1 a_2 - b_1 b_2) - (a_1 b_2 + a_2 b_1)i \quad \text{--- (i)}$$

$$\text{R.H.S} = \overline{z_1 z_2}$$

$$= \overline{(a_1 + b_1 i)(a_2 + b_2 i)}$$

$$= \overline{a_1 a_2 - a_1 b_2 i - a_2 b_1 i + b_1 b_2 i^2}$$

$$= \overline{a_1 a_2 - a_1 b_2 i - a_2 b_1 i + b_1 b_2 (-1)}$$

$$= \overline{a_1 a_2 - a_1 b_2 i - a_2 b_1 i - b_1 b_2}$$

$$= \overline{a_1 a_2 - b_1 b_2 - a_1 b_2 i - a_2 b_1 i}$$

$$= (a_1 a_2 - b_1 b_2) - (a_1 b_2 + a_2 b_1)i \quad \text{--- (ii)}$$

From (i) and (ii)

$$\text{L.H.S} = \text{R.H.S}$$

07. Find the square root of the following complex numbers:

(i) $-7 - 24i$

Sol. By using the square root formula for complex numbers

$$\sqrt{x+iy} = \pm \left(\sqrt{\frac{x^2+y^2}{2}} + i \sqrt{\frac{y^2-x^2}{2}} \right)$$

$$\sqrt{-7-24i} =$$

$$\pm \left(\sqrt{\frac{(-7)^2 + (-24)^2}{2}} + i \sqrt{\frac{(-24)^2 - (-7)^2}{2}} \right)$$

$$\sqrt{-7-24i} = \pm \left(\sqrt{\frac{49+576}{2}} - i \sqrt{\frac{576-49}{2}} \right)$$

$$\sqrt{-7-24i} = \pm \left(\sqrt{\frac{625}{2}} - i \sqrt{\frac{625}{2}} \right)$$

$$\sqrt{-7-24i} = \pm \left(\frac{25}{\sqrt{2}} - i \sqrt{\frac{25}{2}} \right)$$

$$\sqrt{-7-24i} = \pm \left(\frac{18}{2} - i \sqrt{\frac{32}{2}} \right)$$

$$\sqrt{-7-24i} = \pm (\sqrt{9} - i\sqrt{16})$$

Either

$$\sqrt{-7-24i} = + (3 - 4i) \quad \text{or} \quad \sqrt{-7-24i} = - (3 - 4i)$$

$$\sqrt{-7-24i} = 3 - 4i \quad \text{or} \quad \sqrt{-7-24i} = -3 + 4i$$

(ii) $8 - 6i$

Sol. By using the square root formula for complex numbers

$$\sqrt{x+iy} = \pm \left(\sqrt{\frac{x^2+y^2}{2}} + i \sqrt{\frac{y^2-x^2}{2}} \right)$$

$$\sqrt{8-6i} =$$

$$\pm \left(\sqrt{\frac{(8)^2 + (-6)^2}{2}} + i \sqrt{\frac{(-6)^2 - (8)^2}{2}} \right)$$

$$\sqrt{8-6i} = \pm \left(\sqrt{\frac{64+36}{2}} - i \sqrt{\frac{36-64}{2}} \right)$$

$$\sqrt{8-6i} = \pm \left(\sqrt{\frac{100}{2}} - i \sqrt{\frac{100-8}{2}} \right)$$

$$\sqrt{8-6i} = \pm \left(\sqrt{\frac{10+8}{2}} - i \sqrt{\frac{10-8}{2}} \right)$$

$$\sqrt{8-6i} = \pm \left(\sqrt{\frac{18}{2}} - i \sqrt{\frac{2}{2}} \right)$$

$$\sqrt{8-6i} = \pm (\sqrt{9} - i\sqrt{1})$$

Either

$$\sqrt{8-6i} = + (3 - i) \quad \text{or} \quad \sqrt{8-6i} = - (3 - i)$$

$$\sqrt{8-6i} = 3 - i \quad \text{or} \quad \sqrt{8-6i} = -3 + i$$

(iii) $-15 - 36i$

Sol. By using the square root formula for complex numbers

$$\sqrt{x+iy} = \pm \left(\sqrt{\frac{x^2+y^2}{2}} + i \sqrt{\frac{y^2-x^2}{2}} \right)$$

$$\sqrt{-15-36i} = \pm$$

$$\pm \left(\sqrt{\frac{(-15)^2 + (-36)^2}{2}} + i \sqrt{\frac{(-36)^2 - (-15)^2}{2}} \right)$$

$$\sqrt{-15-36i} = \pm \left(\sqrt{\frac{225+1296}{2}} - i \sqrt{\frac{1296-225}{2}} \right)$$

$$\sqrt{-15-36i} = \pm \left(\sqrt{\frac{1521-15}{2}} - i \sqrt{\frac{1521+15}{2}} \right)$$

$$\sqrt{-15-36i} = \pm \left(\sqrt{\frac{39-15}{2}} - i \sqrt{\frac{39+15}{2}} \right)$$

$$\sqrt{-15-36i} = \pm \left(\sqrt{\frac{24}{2}} - i \sqrt{\frac{54}{2}} \right)$$

$$\sqrt{-15-36i} = \pm (\sqrt{12} - i\sqrt{27})$$

$$\sqrt{-15-36i} = \pm (2\sqrt{3} - 3\sqrt{3}i)$$

Either $\sqrt{-15-36i} = (2\sqrt{3}-3\sqrt{3}i)$ or $\sqrt{-15-36i} = -(2\sqrt{3}-3\sqrt{3}i)$
 (iv) $119 + 120i$
 Sol. By using the square root formula for complex numbers

$$\sqrt{x+iy} = \pm \left(\sqrt{\frac{x^2+y^2}{2}} + \frac{iy}{|y|} \sqrt{\frac{x^2+y^2}{2}} - x \right)$$

$$\sqrt{119+120i} = \pm \left(\sqrt{\frac{(119)^2+(120)^2}{2}} + \frac{i(120)}{|120|} \sqrt{\frac{(119)^2+(120)^2}{2}} - (119) \right)$$

$$\sqrt{119+120i} = \pm \left(\sqrt{\frac{14161+14400}{2}} + \frac{120i}{120} \sqrt{\frac{14161+14400}{2}} - 119 \right)$$

$$\sqrt{119+120i} = \pm \left(\sqrt{\frac{28561+119}{2}} + i \sqrt{\frac{28561-119}{2}} \right)$$

$$\sqrt{119+120i} = \pm \left(\sqrt{\frac{288}{2}} + i \sqrt{\frac{50}{2}} \right)$$

$$\sqrt{119+120i} = \pm (\sqrt{144} + i\sqrt{25})$$

$$\sqrt{119+120i} = \pm (12+5i)$$

Either

$$\sqrt{119+120i} = (12+5i) \text{ or } \sqrt{119+120i} = -(12+5i)$$

$$\sqrt{119+120i} = 12+5i \text{ or } \sqrt{119+120i} = -12-5i$$

Q8. Find the square root of $13-20\sqrt{3}i$ and represent them on an Argand diagram.

Sol. By using the square root formula for complex numbers

$$\sqrt{x+iy} = \pm \left(\sqrt{\frac{x^2+y^2}{2}} + \frac{iy}{|y|} \sqrt{\frac{x^2+y^2}{2}} - x \right)$$

$$\sqrt{13-20\sqrt{3}i} = \pm \left(\sqrt{\frac{(13)^2+(-20\sqrt{3})^2}{2}} + \frac{i(-20\sqrt{3})}{|-20\sqrt{3}|} \sqrt{\frac{(13)^2+(-20\sqrt{3})^2}{2}} - (13) \right)$$

$$\sqrt{13-20\sqrt{3}i} = \pm \left(\sqrt{\frac{13^2+1200}{2}} - i \sqrt{\frac{13^2+1200}{2}} - 13 \right)$$

$$\sqrt{\frac{169+1200+13}{2} - \frac{20\sqrt{3}i}{20\sqrt{3}}} = \sqrt{\frac{169+1200-13}{2}}$$

$$\sqrt{13-20\sqrt{3}i} = \pm \left(\sqrt{\frac{1369+13}{2}} - i \sqrt{\frac{1369-13}{2}} \right)$$

$$\sqrt{13-20\sqrt{3}i} = \pm \left(\sqrt{\frac{37+13}{2}} - i \sqrt{\frac{37-13}{2}} \right)$$

$$\sqrt{13-20\sqrt{3}i} = \pm \left(\sqrt{\frac{50}{2}} - i \sqrt{\frac{24}{2}} \right)$$

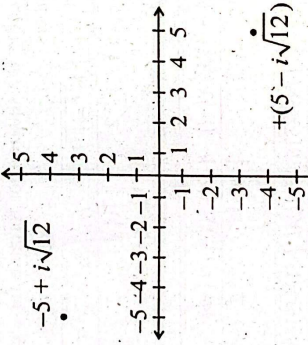
$$\sqrt{13-20\sqrt{3}i} = \pm (\sqrt{25} - i\sqrt{12})$$

$$\sqrt{13-20\sqrt{3}i} = \pm (5-2\sqrt{3}i)$$

Either

$$\sqrt{13-20\sqrt{3}i} = (5-2\sqrt{3}i) \text{ or } \sqrt{13-20\sqrt{3}i} = -(5-2\sqrt{3}i)$$

$$\sqrt{13-20\sqrt{3}i} = 5-2\sqrt{3}i \text{ or } \sqrt{13-20\sqrt{3}i} = -5+2\sqrt{3}i$$



Q9. Find the value of x and y if

$$(-7+i)(x+iy) + (-1-5i) = i(11-i)$$

$$\text{Sol. } (-7+i)(x+iy) - 1-5i = 11i-i^2$$

$$(-7+i)(x+iy) - 1-5i = 11i-(-1)$$

$$(-7+i)(x+iy) - 1-5i = 11i+1$$

$$(-7+i)(x+iy) = 11i+1+1+5i$$

$$(-7+i)(x+iy) = 2+16i$$

$$x+iy = \frac{2+16i}{-7+i}$$

$$x+iy = \frac{2+16i}{-7+i} \times \frac{-7-i}{-7-i}$$

$$x+iy = \frac{-14-2i-112i-16i^2}{49-(-1)}$$

$$x+iy = \frac{-14-2i-112i-16(-1)}{49+1}$$

$$x+iy = \frac{-14-2i-112i+16}{49+1}$$

$$x+iy = \frac{2-114i}{50}$$

$$x+iy = \frac{2}{50} - \frac{114i}{50}$$

$$x+iy = \frac{1}{25} - \frac{57i}{25}$$

$$x = \frac{1}{25} \text{ and } y = -\frac{57}{25}$$

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$$x+iy = \frac{2-114i}{50}$$

$$x+iy = \frac{2}{50} - \frac{114i}{50}$$

$$x+iy = \frac{1}{25} - \frac{57i}{25}$$

By comparing the real and imaginary parts

$$x = \frac{1}{25} \text{ and } y = -\frac{57}{25}$$

Q10. Find the value of x and y if

$$(5-2i)(x+iy) + 3 = i(11-i) - 4i$$

$$\text{Sol. } (5-2i)(x+iy) + 3 = 11i-i^2-4i$$

$$(5-2i)(x+iy) + 3 = 11i-(-1)-4i$$

$$(5-2i)(x+iy) = 11i+1-4i-3$$

$$(5-2i)(x+iy) = -2+7i$$

$$x+iy = \frac{-2+7i}{5-2i}$$

$$x+iy = \frac{-2+7i}{5-2i} \times \frac{5+2i}{5+2i}$$

$$x+iy = \frac{-10-4i+35i+14i^2}{25-4i^2}$$

$$x+iy = \frac{-10-4i+35i+14(-1)}{25-4(-1)}$$

$$x+iy = \frac{-10-4i+35i+14(-1)}{25-4(-1)}$$

$$x+iy = \frac{-10-4i+35i-14}{25+4}$$

$$x+iy = \frac{-24+31i}{29}$$

$$x+iy = \frac{24}{29} + \frac{31i}{29}$$

$$x = \frac{24}{29} \text{ and } y = \frac{31}{29}$$

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Q12. If $z_1 = 4 + 5i$ and $z_2 = \alpha - 2i$, find the value of α such that $\text{Re}(z_1 z_2) = 20$.

Sol. For $z_1 z_2$

$$z_1 z_2 = (4 + 5i)(\alpha - 2i)$$

$$z_1 z_2 = 4\alpha - 8i + 5\alpha i - 10i^2$$

$$z_1 z_2 = 4\alpha - 8i + 5\alpha i - 10(-1)$$

$$z_1 z_2 = 4\alpha - 8i + 5\alpha i + 10$$

$$z_1 z_2 = 4\alpha + 10 - 8i + 5\alpha i$$

$$z_1 z_2 = (4\alpha + 10) - (8 - 5\alpha)i$$

$$\text{Re}(z_1 z_2) = 4\alpha + 10 \text{ and } \text{Im}(z_1 z_2) = -(8 - 5\alpha)$$

$$\text{Given } \text{Re}(z_1 z_2) = 20$$

$$\text{Put the value of } \text{Re}(z_1 z_2)$$

$$4\alpha + 10 = 20$$

$$4\alpha = 20 - 10$$

$$4\alpha = 10$$

$$\alpha = \frac{10}{4} \Rightarrow \alpha = \frac{5}{2}$$